

Unit 3 Introduction to Functions

Learning Goals:

- explain the difference between a function and a relation;
- demonstrate an understanding of functions, their representations, and their inverses, and make connections between the algebraic and graphical representations of functions using transformations;
- explain the meaning of the terms domain and range;
- determine, through investigation, the relationship between the domain and range of a function and the domain and range of the inverse relation, and determine whether or not the inverse relation is a function
- determine, through investigation using technology, the roles of the parameters a , k , d , and c in functions of the form
- $y = af(k(x - d)) + c$, and describe these roles in terms of transformations on the graphs of the parent functions.

Self-Assessment

H	R	G
H	R	G
H	R	G
H	R	G
H	R	G
H	R	G

H – I need to ask for **HELP** with this.

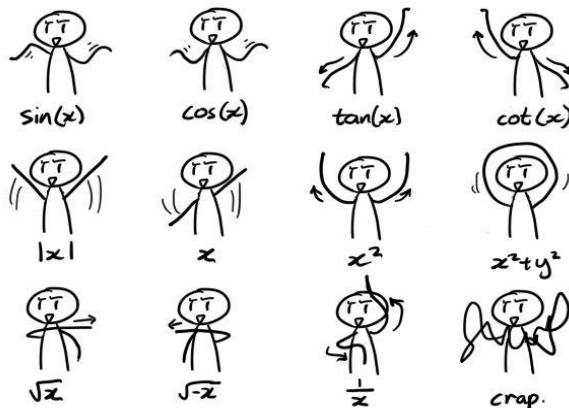
R – I need to **REVIEW** this.

G – I've **GOT THIS** down.

Homework Log

Lesson	Section	Assigned Work	Extra Practice
1	1.1 Functions & Relations 1.4 Domain & Range	Text p. 10-11 #1-3, 6, 9 Text p. 35-37 #1-3, 5, 10, 14, 16	Text: p. 11 #7, 8, 10 Text: p. 37 #8, 13,
2	1.2 Function Notation	LP p. 8 Text p. 22-24 #6, 15, 17, 20	Text p. 22-24 #1 – 5, 7 – 11
3	1.3 Graphing the base (parent) functions	Text p. 28 #1 – 3 LP P. 9 – 10	
4	1.6 Horizontal and Vertical Translations of Functions	LP p. 17 – 18	
5	1.7 Stretches and Reflections of Functions	Text. p. 58-60 #1 – 3, 8, 10 – 12, 14, 15	Text p. 58-60 #4 – 7, 13
6	1.8 Combinations of Transformations	LP p. 26 – 27	Text p. 70-73 #4 – 13
7	1.5 Inverse Functions	LP p. 30 – 31	Text p. 47-48 #2c-f, 4, 10, 11 Text p. 160-161 #2, 3, 7, 8
8	Review for Test	LP pp. 32 – 34	Text p. 76 – 77 #1 – 6, 8, 10, 12 – 18

Mathematical Dance Moves



1.1 Relations and Functions

Relations

In a relation, values of the _____ variable are paired with values of the _____ variable.

A relation can be expressed

- in words
- as a table of values
- as a set of ordered pairs
- as a scatter plot (graph)
- as an equation
- as a mapping diagram

Ex 1: Jason has a summer job cutting lawns. He is paid \$10 per lawn. The amount he earns is related to the number of lawns he cuts. Display the data in the following methods.

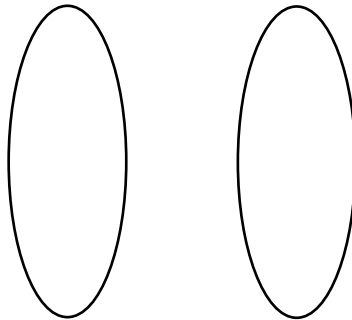
- a table of values



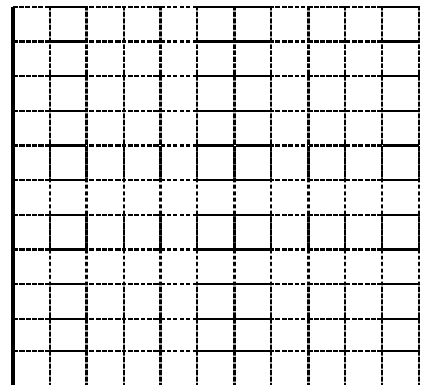
x	y

- a set of ordered pairs

- a mapping diagram



- a scatter plot



- an equation

Defining a Relation as a Function

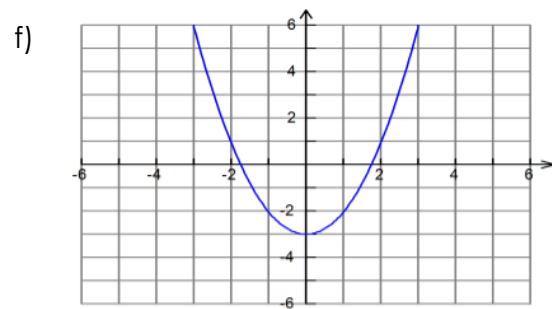
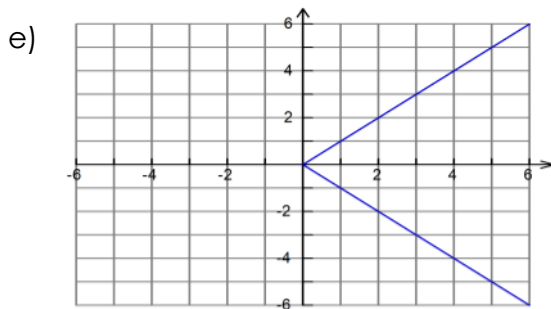
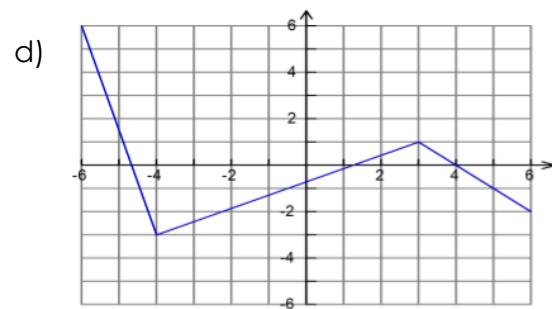
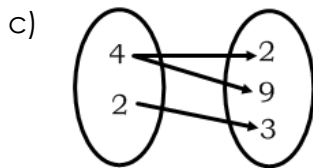
A **function** is a relation (set of ordered pairs) in which, for every ____ (_____), there is **only one** ____ (_____).

An easy way of determining whether or not a relation is a function is to use its graph and the _____.

Ex. 2: Determine if each relation is a function.

a) $A = \{(2, 1), (3, 5), (3, 6), (4, -2), (6, 4)\}$

b) $B = \{(3, -2), (4, 1), (5, 1), (6, 0), (-3, 2)\}$



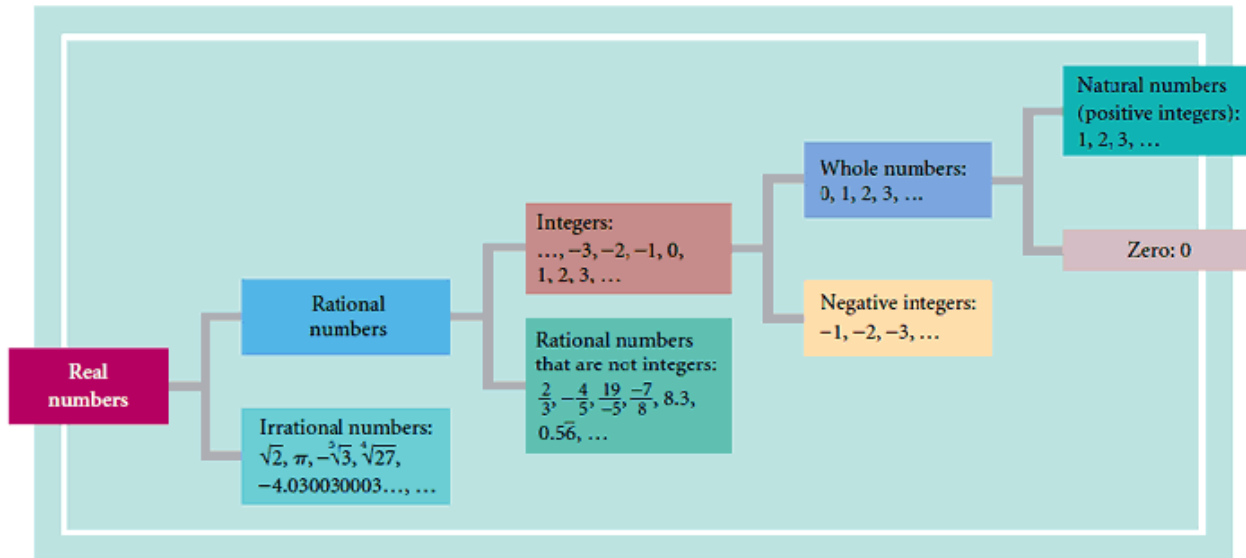
g) $y = \frac{3}{4}x - 9$

Domain and Range

Given a relation, the domain is the set of first elements (x-coordinates, first coordinates) in the ordered pair.

The range is the set of second elements (y-coordinates, second coordinates).

Sets of Real Numbers



(Source: Beecher, Penna & Bittinger, Algebra and Trigonometry, p. 2)

The Notation:

Domain: $\{x \in \mathbb{R} \mid x \neq 0\}$

Range: $\{y \in \mathbb{Z} \mid 0 < y \leq 10\}$

Ex 3. For each of the following:

- State the domain;
- State the range;
- State whether the relation is a function.

a) $A = \{(0, -2), (0, -1), (0, 0), (0, 5)\}$

Domain: _____

Range: _____

Function? _____

b)

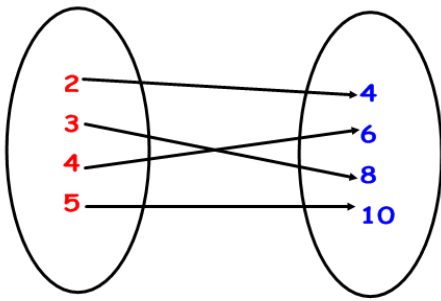
x	y
-1	-3
0	1
1	5
2	9

Domain: _____

Range: _____

Function? _____

c)

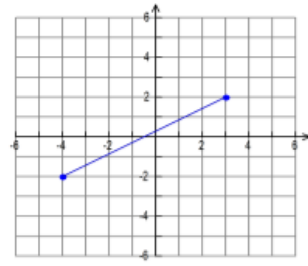


Domain: _____

Range: _____

Function? _____

d)

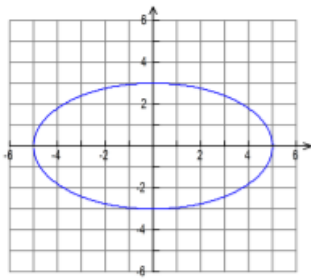


Domain: _____

Range: _____

Function? _____

e)

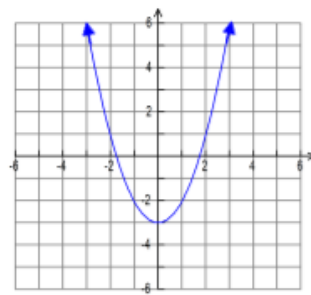


Domain: _____

Range: _____

Function? _____

f)

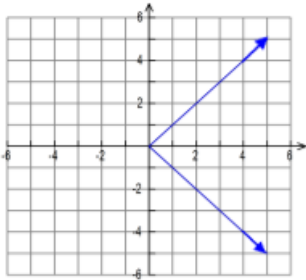


Domain: _____

Range: _____

Function? _____

g)



Domain: _____

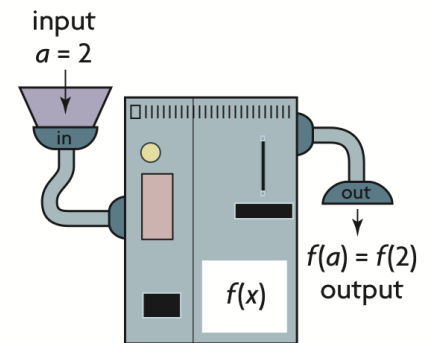
Range: _____

Function? _____

1.2 Function Notation

Recall, relations can be represented in equation form. The equation can be written in x-y notation as $y = -x^2 + 8x$ or $x^2 + y^2 = 25$. Relations that are functions can be written in function notation as $f(x) = -x^2 + 8x$. The notations y and $f(x)$ are interchangeable in the equation or graph of a function, so y is equal to $f(x)$. The symbols $f(x)$, $g(x)$, and $h(x)$ are often used to name the outputs of functions, but other letters are also used, such as $v(t)$ for velocity as a function of time.

- The notation $f(x)$ is read "f at x" or "f of x"
- $f(x)$ is used to represent the value of the dependent variable; it is another name for y
- $f(a)$ represents the value or output of the function when the input is $x = a$. The output depends on the equation of the function. To evaluate $f(a)$, substitute a for x in the equation for $f(x)$.
- $f(a)$ is the y-coordinate of the point on the graph of f with x-coordinate a . For example, if $f(x)$ takes the value 3 at $x = 2$, then $f(2) = 3$ and the point $(2, 3)$ lies on the graph of f .



Ex. 1: If $H(t) = 3t^2 + 5t - 11$, find each of the following:

a) $H(1/2)$

b) $H(4 - 1)$

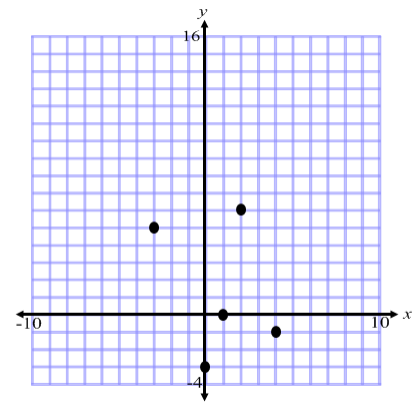
c) $H(3) + H(2)$

d) $H(m)$

e) $H(3x)$

Ex. 2: Consider the functions $f(x) = x^2 - 3x$ and $g(x) = 1 - 2x$. Determine $f(c + 2) - g(c + 2)$.

Ex. 3: The graph of $y = f(x)$ is shown below.



a) State the domain and range of $f(x)$.

b) Evaluate:

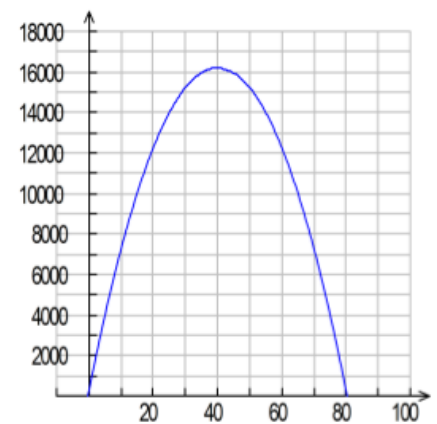
i) $f(4) =$

ii) $f(1 + 1) =$

iii) $f(1) + f(0) =$

c) $f(-3) = 5$. What is the corresponding ordered pair? What does -3 represent, and what does $f(-3)$ represent?

Ex. 4: The relationship between the selling price of a pair of sunglasses and revenue, $R(s)$, is represented by the function: $R(s) = -10s^2 + 800s + 120$ and its graph is below.



a) Identify the independent variable and the dependent variable. Label each one appropriately, on its respective axis of the graph.

b) Determine the revenue when the selling price is \$5.

c) Explain what $R(20) = 12120$ means.

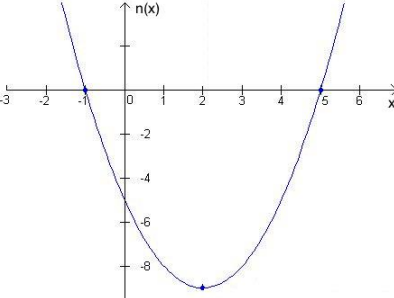
d) If $R(s) = 16120$, determine the selling price s using the graph or the equation.

Function Notation Practice

Using the functions below, determine the following values:

1. $f(2)$	2. $g(3)$
3. $h(-4)$	4. $k(2)$
5. $n(5)$	6. $2m(-4) - m(3)$
7. $f(3) + n(2)$	8. $k(x)$
9. $h(2a + 1)$	10. $g(5) \cdot m(-1)$
11. $k(x - 1) \cdot f(1)$	12. $f(m(4))$
13. $h(a+1) - h(a)$	14. $k(h(x))$

FUNCTIONS TO USE

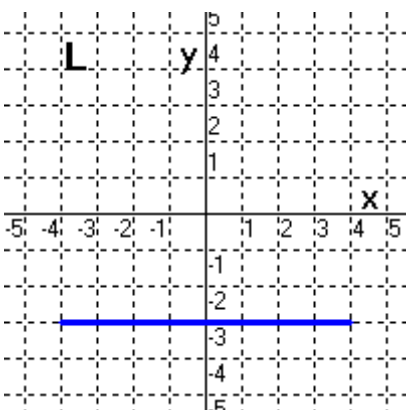
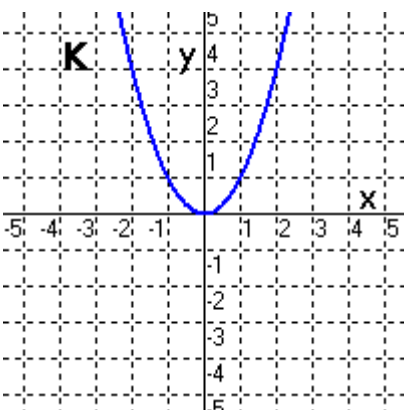
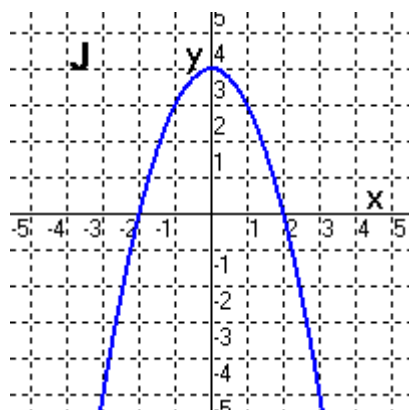
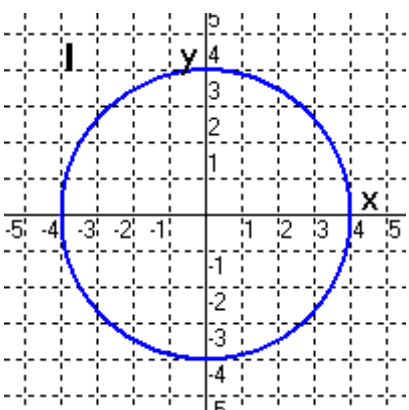
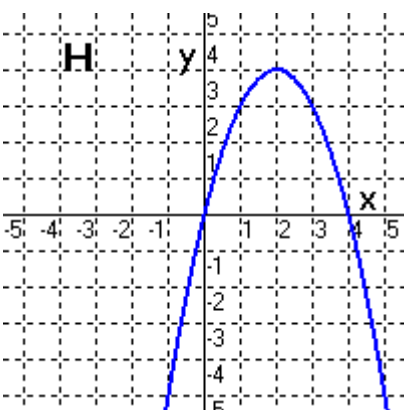
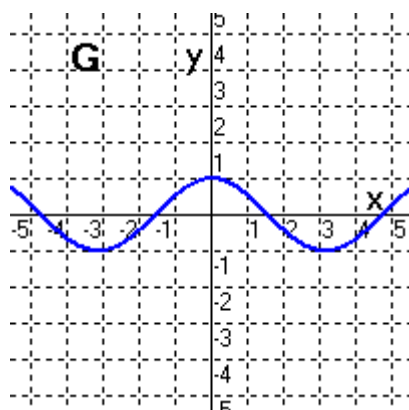
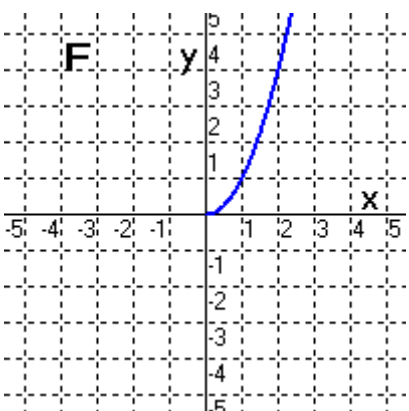
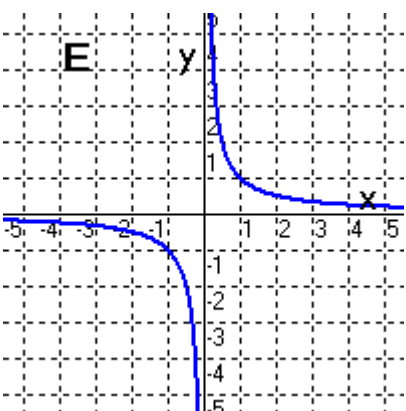
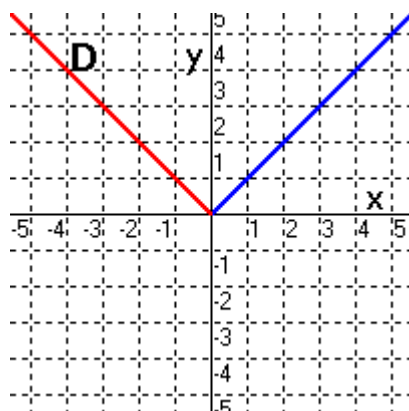
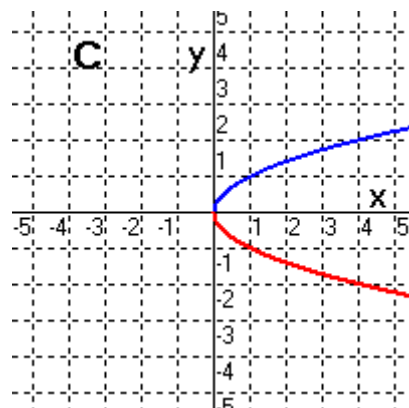
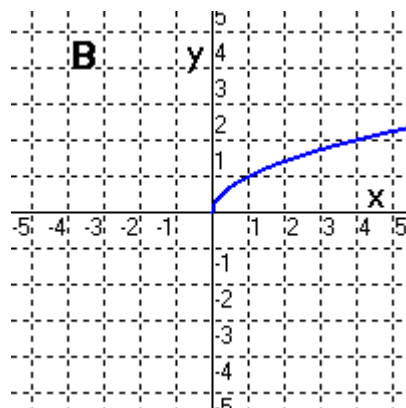
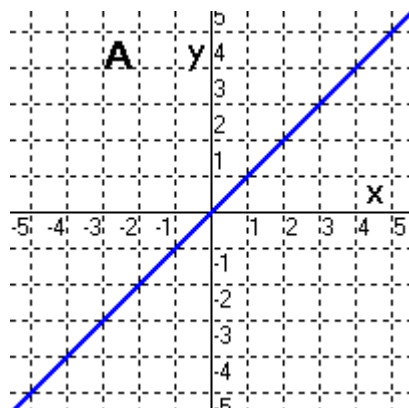
		$h(x) = 2x + 1$	$f = \{(1, 2), (2, 5), (3, 0)\}$																											
$x \longrightarrow$	Triple the number	$\longrightarrow$	Subtract from 10	$\longrightarrow$	Multiple by original number	$\longrightarrow$	$k(x)$																							
<table><tr><th>x</th><th>$g(x)$</th></tr><tr><td>3</td><td>-2</td></tr><tr><td>5</td><td>7</td></tr><tr><td>-4</td><td>-5</td></tr><tr><td>0</td><td>1</td></tr><tr><td>11</td><td>3</td></tr></table>		x	$g(x)$	3	-2	5	7	-4	-5	0	1	11	3	<table><tr><th>x</th><th>$m(x)$</th></tr><tr><td>-4</td><td>3</td></tr><tr><td>-1</td><td>0</td></tr><tr><td>6</td><td>-4</td></tr><tr><td>3</td><td>1</td></tr><tr><td>4</td><td>-5</td></tr></table>		x	$m(x)$	-4	3	-1	0	6	-4	3	1	4	-5	$n(x)$: 		
x	$g(x)$																													
3	-2																													
5	7																													
-4	-5																													
0	1																													
11	3																													
x	$m(x)$																													
-4	3																													
-1	0																													
6	-4																													
3	1																													
4	-5																													

Answers:

1. 5 2. -2 3. -7 4. 8 5. 0 6. -6 7. -9 8. $x(10 - 3x)$ 9. $4a + 3$ 10. 7 11. $2(x - 1)(13 - 3x)$
 12. 0 13. 2 14. $(2x + 1)(7 - 6x)$

MCR3U- Functions or Relations?

Part A : State which of the following are functions, and which are relations.

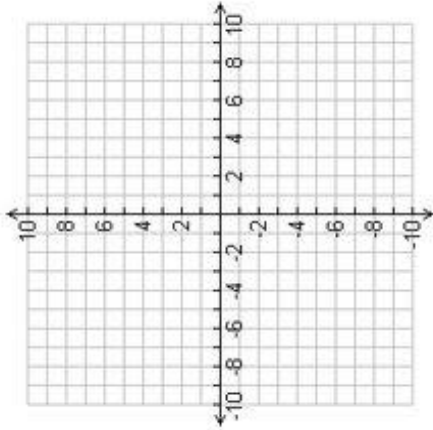
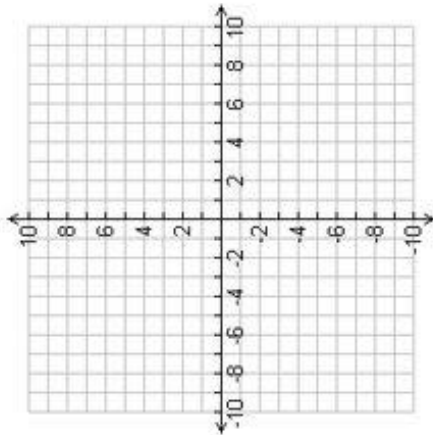


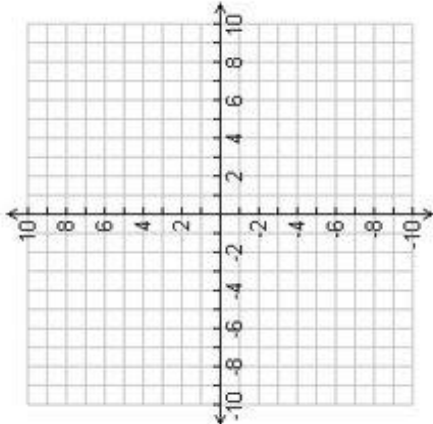
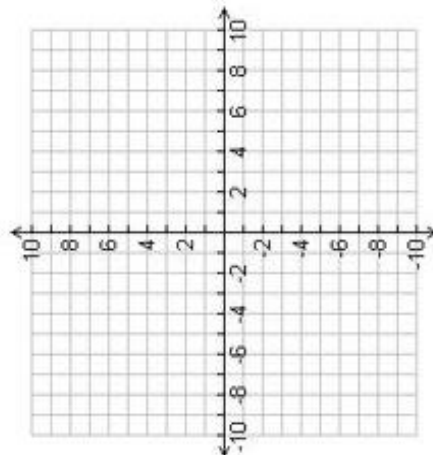
Part B : Complete the table:

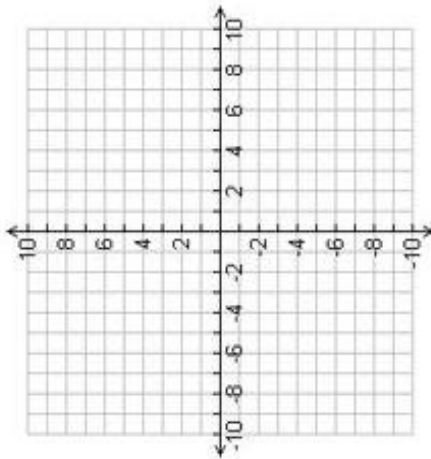
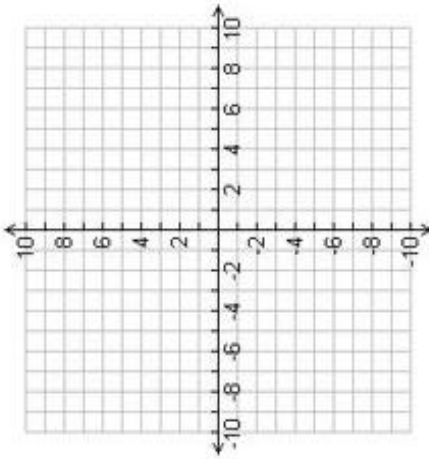
Graph	Function or relation?	Domain	Range	Equation (state if known, guess if not)
A				
B				
C				
D				
E				
F				
G				
H				
I				
J				
K				
L				

1.3 Exploring Properties of Parent Functions

In mathematics, every function can be classified as a member of a **family**. Each member of a family of functions is related to the simplest, or most basic, function sharing the same characteristics. This function is called the **parent function**.

Function	Table of Values	Sketch of Graph	Domain & Range	Special Features/Symmetry												
Linear Function $f(x) = x$	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															
Quadratic Function $f(x) = x^2$	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															

Function	Table of Values	Sketch of Graph	Domain & Range	Special Features/Symmetry												
Cubic Function $f(x) = x^3$	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															
Root Function $f(x) = \sqrt{x}$	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															

Function	Table of Values	Sketch of Graph	Domain & Range	Special Features/Symmetry												
Reciprocal Function $f(x) = \frac{1}{x}$	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															
Absolute Value Function $f(x) = x $	<table><tr><td>x</td><td>y</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	x	y													
x	y															

1.6 Exploring Transformations of Parent Functions – Horizontal and Vertical Translations

Recall in Grade 10, we used transformations to change the parabola. We can apply these same transformations to all of the parent functions. For the parabola, we were able to use the step property to graph transformed parabolas, however that will not be the case for other parent functions. Instead, we will use **Mapping Notation** to graph transformed functions.

Login in to D2L, to access the accompanying activity for this lesson.

A. Vertical Translations, $g(x) = f(x) \pm c$

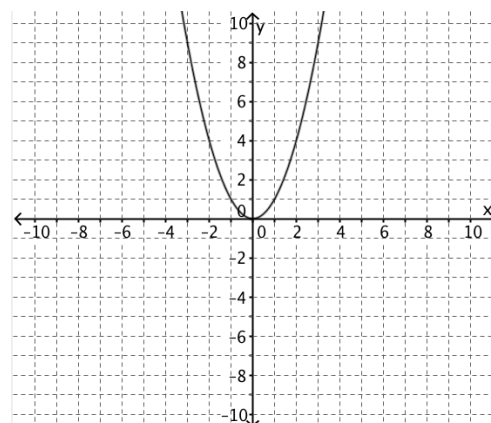
Use the k slider to transform the functions, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \frac{1}{x}$ and $f(x) = |x|$. Note: to graph the square root function, type $\rightarrow \text{sqrt}(x)$ and to graph the absolute value function, type $\rightarrow \text{abs}(x)$.

1. What is the effect of adding a constant c to $f(x)$? What is the effect of subtracting a constant c to $f(x)$?
2. For the reciprocal function, $f(x) = \frac{1}{x}$, how does applying a vertical translation affect the asymptotes?

We can use the transformations to determine the coordinates of transformed functions. This is called **Mapping Notation**. Vertical translations affect the y-coordinate, therefore, the mapping notation will be $(x, y \pm c)$.

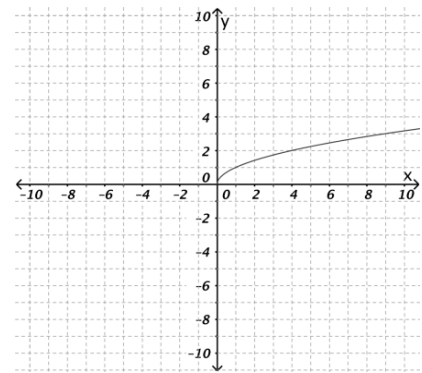
Ex. 1: Graph the following functions using mapping notation. State the domain and range for each function.

a) $f(x) = x^2 - 7$



Domain: _____ Range: _____

b) $f(x) = \sqrt{x} + 3$



Domain: _____ Range: _____

B. Horizontal Translations, $g(x) = f(x \pm d)$

Use the h slider to transform the functions, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \frac{1}{x}$ and $f(x) = |x|$.

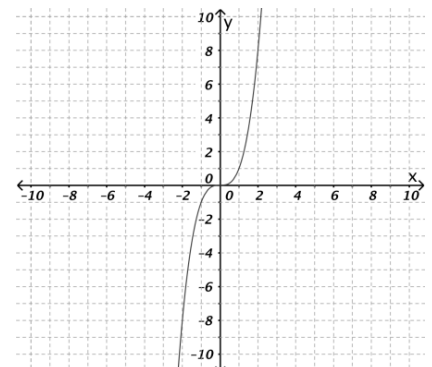
1. What is the effect of adding a constant h to $f(x)$? What is the effect of subtracting a constant h to $f(x)$?

2. For the reciprocal function, $f(x) = \frac{1}{x}$, how does applying a horizontal translation affect the asymptotes?

Horizontal translations affect the x-coordinate, therefore, the mapping notation will be $(x \pm d, y)$.

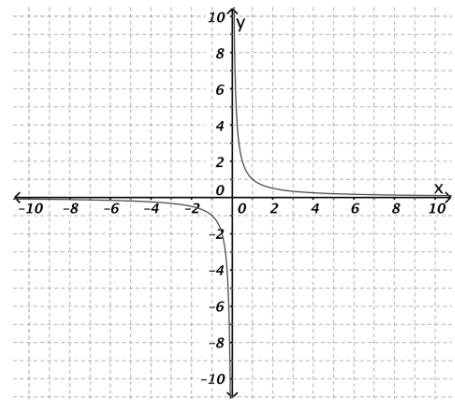
Ex. 2: Graph the following functions using mapping notation. State the domain and range for each function.

a) $f(x) = (x - 5)^3$



Domain: _____ Range: _____

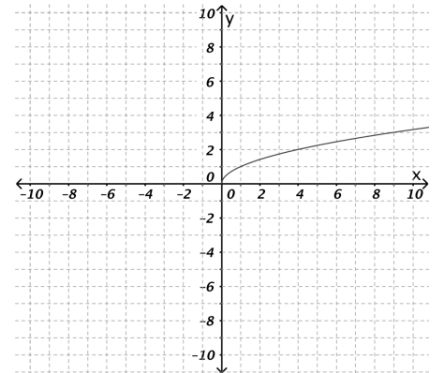
b) $f(x) = \frac{1}{x+4}$



Domain: _____ Range: _____

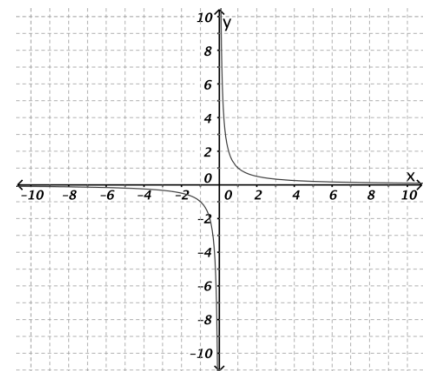
Ex. 3: Graph the following functions using mapping notation. State the domain and range for each function.

a) $f(x) = \sqrt{x+1} - 7$



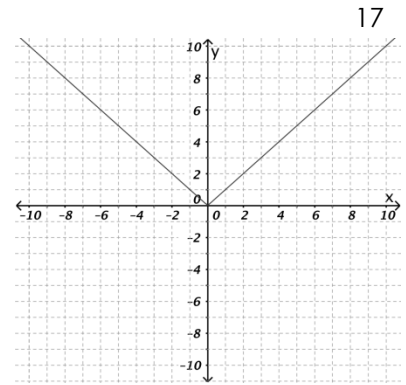
Domain: _____ Range: _____

b) $f(x) = \frac{1}{x-2} + 3$



Domain: _____ Range: _____

c) $f(x) = |x+2| - 8$



Domain: _____ Range: _____

Horizontal and Vertical Translations of Functions Practice

1. The function $y = f(x)$ is given. Describe how the graphs of the following functions can be obtained from the graph $y = f(x)$.

a) $y = f(x) + 5$

b) $y = f(x) - 6$

c) $y = f(x - 4)$

d) $y = f(x + 8)$

e) $y = f(x + 3) - 5$

f) $y = f(x - 6) + 2$

g) $y = f(x - 5) - 7$

h) $y = f(x + 2) + 9$

2. The function $y = f(x)$ has been transformed to $y = f(x - d) + c$. Determine the values of d and c for each of the following transformations.

a) 6 units upward

b) 8 units downward

c) 3 units to the right

d) 5 units to the left

e) 2 units to the left and 4 units downward

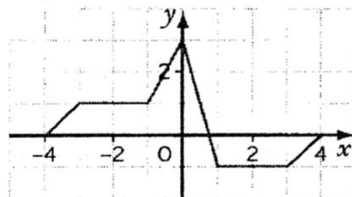
f) 7 units to the right and 7 units upward

3. The graph of a function $y = f(x)$ is shown. Sketch the graph of each of the following.

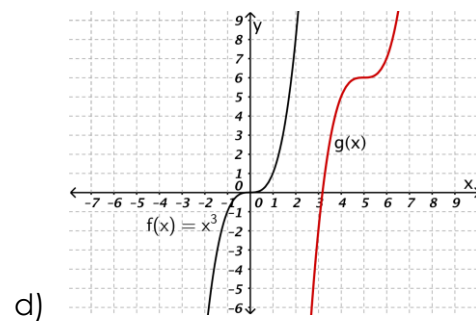
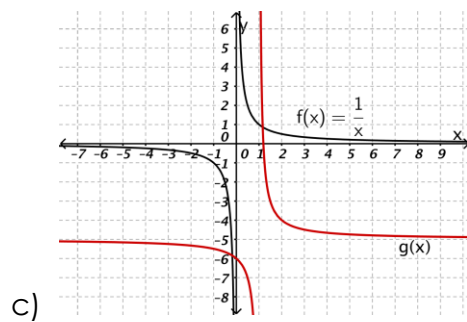
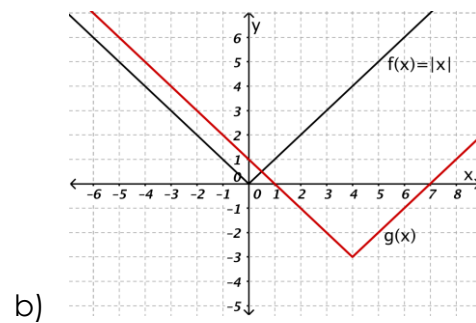
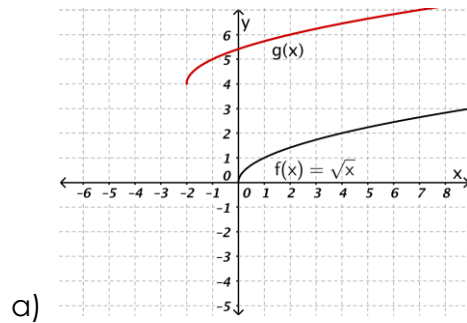
a) $y = f(x) - 4$

b) $y = f(x + 2)$

c) $y = f(x - 3) + 5$



4. The graph of the function, $g(x)$ is a translated image of the function, $f(x)$. Write an equation for each function, $g(x)$.



5. Use mapping notation to sketch the graph of each of the following functions. State the domain and range for each. Graph each family on the same grid.

a) $f(x) = x^2 - 3$

b) $f(x) = (x + 3)^2$

c) $f(x) = (x - 1)^2 + 2$

d) $f(x) = x^3 + 5$

e) $f(x) = (x - 2)^3$

f) $f(x) = (x + 4)^3 - 6$

g) $f(x) = \sqrt{x} - 8$

h) $f(x) = \sqrt{x+1}$

i) $f(x) = \sqrt{x-7} + 4$

j) $f(x) = \frac{1}{x} - 4$

k) $f(x) = \frac{1}{x-6}$

l) $f(x) = \frac{1}{x+1} + 9$

m) $f(x) = |x| + 2$

n) $f(x) = |x + 4|$

o) $f(x) = |x - 6| - 1$

Answers

1. a) translated the graph of $y = f(x)$ upward 5 units b) translated the graph of $y = f(x)$ downward 6 units
 c) translated the graph of $y = f(x)$ to the right 5 units d) translated the graph of $y = f(x)$ to the left 8 units
 e) translated the graph of $y = f(x)$ to the left 3 units and downward 5 units f) translated the graph of $y = f(x)$ to the right 6 units upward 2 units
 g) translated the graph of $y = f(x)$ to the right 5 units and downward 7 units
 h) translated the graph of $y = f(x)$ to the left 2 units and upward 9 units
 2. a) $y = f(x) + 6$ b) $y = f(x) - 8$ c) $y = f(x - 3)$ d) $y = f(x + 5)$ e) $y = f(x + 2) - 4$ f) $y = f(x - 7) + 7$

4. a) $g(x) = \sqrt{x+2} + 4$ b) $g(x) = |x - 4| - 3$ c) $g(x) = \frac{1}{x-1} - 5$ d) $g(x) = (x - 5)^3 + 6$

1.7 Exploring Transformations of Parent Functions – Horizontal and Vertical Stretches and Reflections

Login in to D2L, to access the accompanying activity for this lesson.

A. Vertical Stretches and Reflections, $g(x) = af(x)$

Use the a slider to transform the functions, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \frac{1}{x}$ and

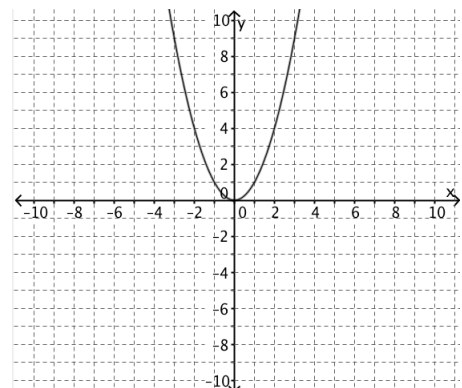
$f(x) = |x|$. Remember, to graph the square root function, type $\rightarrow \text{sqrt}(x)$ and to graph the absolute value function, type $\rightarrow \text{abs}(x)$.

1. What is the effect of multiplying $f(x)$ by a values greater than 1? What is the effect of multiplying $f(x)$ by a values between 0 and 1?
2. What is the effect of multiplying $f(x)$ by negative values of a ?
3. For the reciprocal function, $f(x) = \frac{1}{x}$, how does applying a vertical stretch or reflection affect the asymptotes?

Vertical stretches and reflections affect the y-coordinate, therefore, the mapping notation will be (x, ay) .

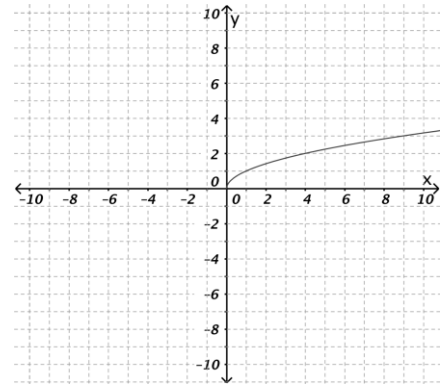
Ex. 1: Graph the following functions using mapping notation. State the domain and range for each function.

a) $f(x) = 3x^2$



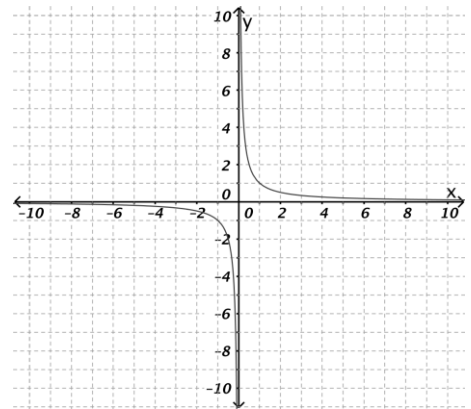
Domain: _____ Range: _____

b) $f(x) = \frac{1}{2}\sqrt{x}$



Domain: _____ Range: _____

c) $f(x) = \frac{-1}{x}$



Domain: _____ Range: _____

B. Horizontal Stretches and Reflections, $g(x) = f(kx)$

Use the b slider to transform the functions, $f(x) = x^2$, $f(x) = x^3$, $f(x) = \sqrt{x}$, $f(x) = \frac{1}{x}$ and

$f(x) = |x|$.

1. What is the effect of multiplying the x-coordinate of $f(x)$ by k values greater than 1?
What is the effect of multiplying the x-coordinate of $f(x)$ by k values between 0 and 1?

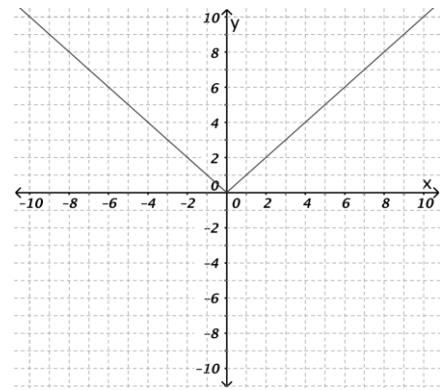
2. What is the effect of multiplying the x-coordinate of $f(x)$ by negative values of k?

3. For the reciprocal function, $f(x) = \frac{1}{x}$, how does applying a horizontal stretch or reflection affect the asymptotes?

Horizontal stretches and reflections affect the x-coordinate, therefore, the mapping notation will be $\left(\frac{1}{k}x, y\right)$.

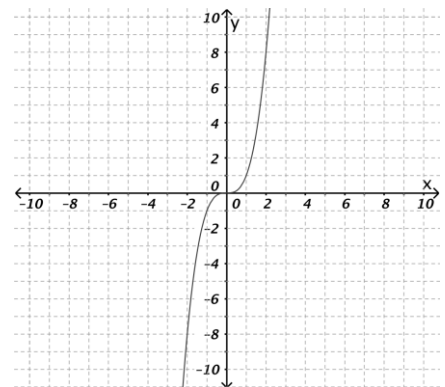
Ex. 2: Graph the following functions using mapping notation. State the domain and range for each function.

a) $f(x) = |2x|$



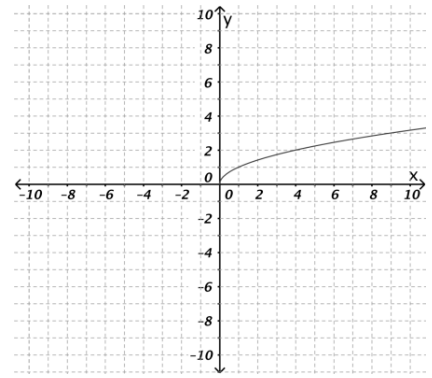
Domain: _____ Range: _____

b) $f(x) = \left(\frac{1}{3}x\right)^3$



Domain: _____ Range: _____

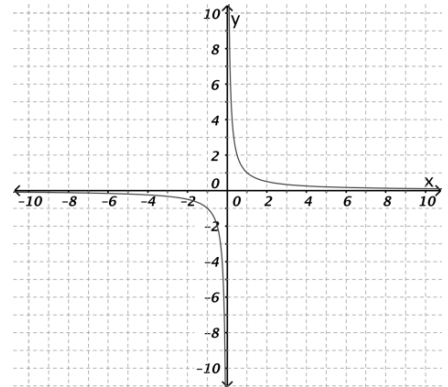
c) $f(x) = \sqrt{-x}$



Domain: _____ Range: _____

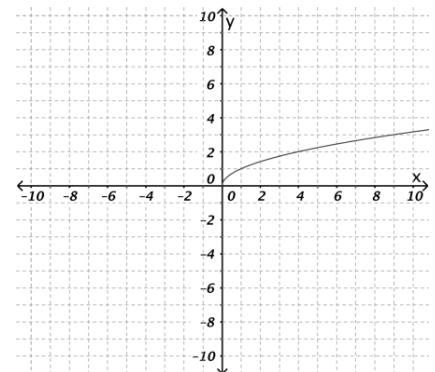
Ex. 3: Graph the following functions using mapping notation. State the domain and range for each function.

d) $f(x) = \frac{-5}{2x}$



Domain: _____ Range: _____

b) $f(x) = \frac{-1}{2} \sqrt{\frac{-1}{3}x}$



Domain: _____ Range: _____

1.8 Using Transformations to Graph Functions of the Form

$$y = af[k(x - d)] + c$$

Perform combinations of transformations in the following order

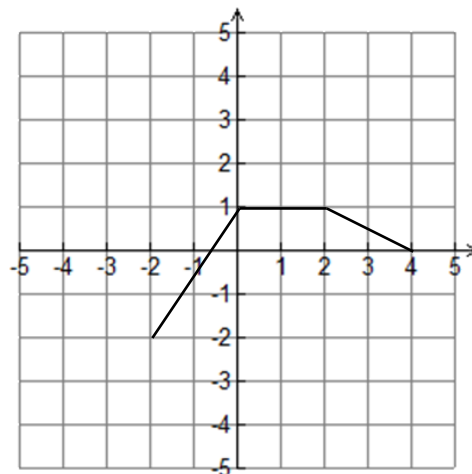
- expansions and compressions
- reflections
- translations

The mapping notation will be $\left(\frac{1}{k}x \pm d, ay \pm c\right)$. If necessary, factor the coefficient of the x-term to identify characteristics of a function more easily.

Ex 1: a) State the mapping notation for $g(x) = 3f(4x - 8)$.

b) If $(-3, 7)$ is a point on $f(x)$, determine the point on $g(x)$.

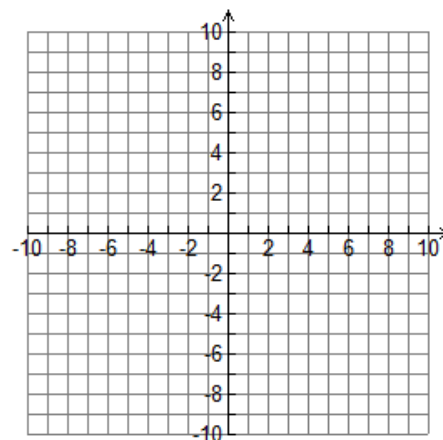
Ex. 2: The graph of $y = f(x)$ is shown. Describe the transformations of $y = -f[2(x + 1)] + 3$. Sketch the graph on the same axes as $y = f(x)$. State the domain and range of the functions.



Domain: _____ Range: _____

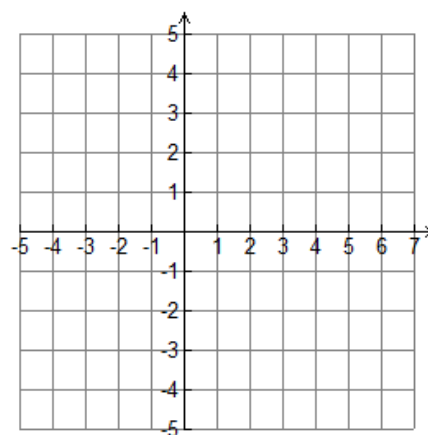
Ex. 3: For each of the following transformed functions, describe the transformations, graph and state the domain and range.

a) $f(x) = 2(x + 5)^2 - 4$



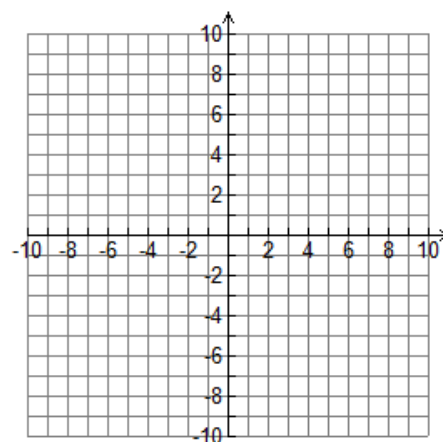
Domain: _____ Range: _____

b) $h(x) = \frac{1}{2}\sqrt{2x - 2}$



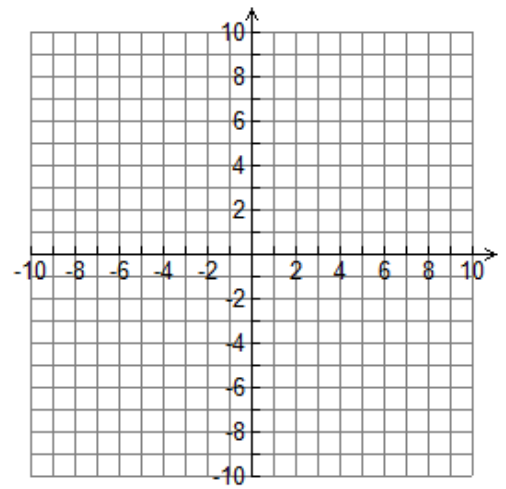
Domain: _____ Range: _____

c) $g(x) = 2|-(x - 3)| + 5$



Domain: _____ Range: _____

d) $h(x) = \frac{-2}{x+1} + 3$



Domain: _____ Range: _____

Practise

- Describe the transformations you would need to apply to the graph of $y = f(x)$ to graph each function.
 - $y = 3f(x) - 1$
 - $y = -f\left(\frac{1}{2}x\right) - 2$
 - $y = f\left(\frac{1}{3}(x+4)\right)$
 - $y = \frac{2}{3}f(x+3) + 1$
 - $y = -3f(2x-2) - 3$
 - $y = f(4-x) + 7$
 - $y = -f(3x+9) - 5$
 - $y = \frac{1}{2}f\left(\frac{1}{3}x\right) + 1$
- If the point $(-1, 5)$ is on $y = f(x)$, determine the new points for each transformed function in question 1.
- Write the equation of the new functions below.
 - The graph of $f(x) = \sqrt{x}$ is reflected across the y-axis, stretched vertically by the factor 3, and then translated 5 units right and 2 units down.
 - The graph of $f(x) = \frac{1}{x}$ is reflected across the x-axis, compressed horizontally by the factor 2, and then translated 1 unit left and 6 units up.
 - The graph of $f(x) = x^3$ is compressed vertically by the factor 4, stretched horizontally by a factor of 3 and then translated 6 units right.
 - The graph of $f(x) = |x|$ is reflected across the y-axis, and then translated 3 units left and 1 unit down.
- Graph the following functions and state the domain and range for each.
 - $f(x) = 2(x+1)^2 - 7$
 - $f(x) = \frac{1}{4}\sqrt{2-x} + 3$
 - $f(x) = \frac{2}{3x-12} - 1$
 - $f(x) = -\left(\frac{1}{2}x+1\right)^3$
 - $f(x) = |-x+5| - 8$
 - $f(x) = \frac{1}{4}x^2 - 9$
 - $f(x) = -\sqrt{-x-4} + 5$
 - $f(x) = \frac{-1}{\frac{1}{4}x+1}$
 - $f(x) = (2x-6)^3$
 - $f(x) = -\frac{1}{2}|x+3| + 7$
- If $f(x) = (x-2)(x+5)$, determine the x-intercepts for each function.
 - $y = f(x)$
 - $y = -4f(x)$
 - $y = f\left(-\frac{1}{3}x\right)$
 - $y = f(-(x+2))$
- A function $f(x)$ has domain $\{x \in \mathbb{R} \mid x^3 - 4\}$ and range $\{y \in \mathbb{R} \mid y < -1\}$. Determine the domain and range of each function.
 - $y = 2f(x)$
 - $y = f(-x)$
 - $y = 3f(x+1) + 4$
 - $y = -2f(-x+5) + 1$

7. Match each equation to its graph. Explain your reasoning.

a) $y = \frac{3}{-(x-2)} + 1$

b) $y = 2|x-3| - 2$

c) $y = -2\sqrt{x+3} - 2$

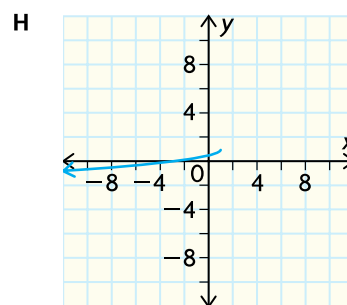
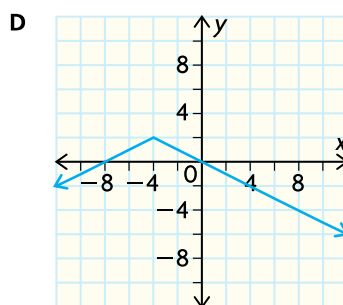
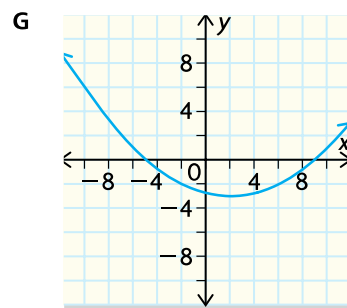
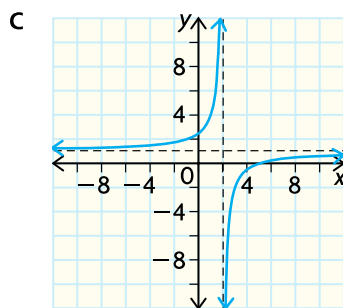
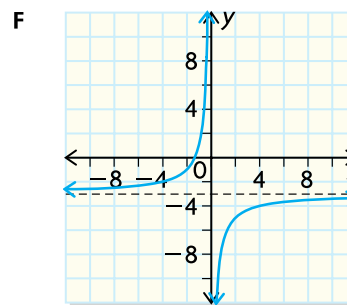
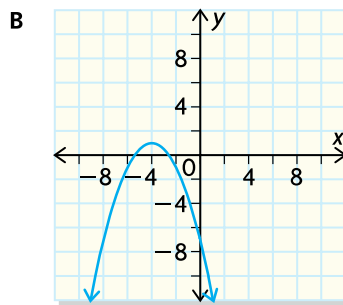
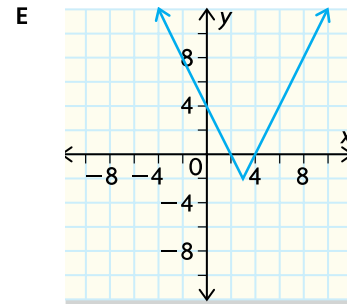
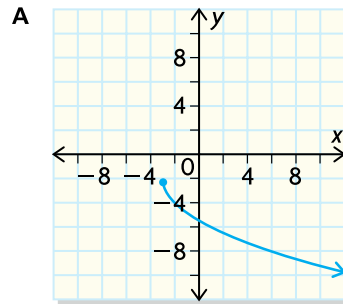
d) $y = \frac{-4}{x} - 3$

e) $y = (0.25(x-2))^2 - 3$

f) $y = -0.5|x+4| + 2$

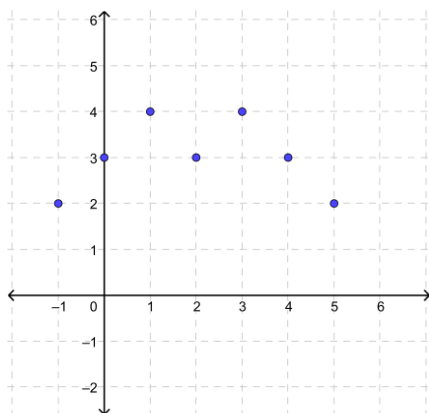
g) $y = -0.5\sqrt{1-x} + 1$

h) $y = \frac{-1}{2}(x+4)^2 + 1$



1.5 The Inverse Function and Its Properties

The following is the graph of a function, $f(x) = \{(-1, 2), (0, 3), (1, 4), (2, 3), (3, 4), (4, 3), (5, 2)\}$.



$$f(x) = \{(-1, 2), (0, 3), (1, 4), (2, 3), (3, 4), (4, 3), (5, 2)\}$$

$$f^{-1}(x) = \{(,), (,), (,), (,), (,), (,), (,)\}$$

Note: The function and its inverse are **symmetric** along the line of symmetry $y = x$.

The inverse of a function is the set formed by **interchanging** each x-coordinate with its corresponding y-coordinate.

That is, the inverse of a function can be found by interchanging the domain and range of the function.

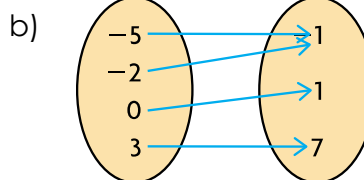
The **inverse of a function** is represented by the notation $f^{-1}(x)$, read “ f inverse” or “the inverse of f .”

$$\text{If } (x, y) \in f(x), \text{ then } (y, x) \in f^{-1}(x)$$

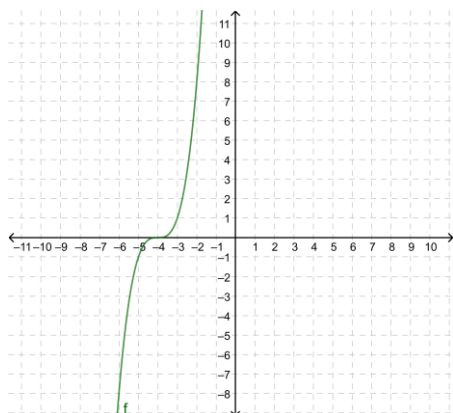
Note: The -1 in f^{-1} is **not** an exponent, so f^{-1} does not mean $\frac{1}{f}$. Also, the inverse of a function is not automatically a function itself. To be a function, it needs to pass the vertical line test.

Ex. 1: Find the inverse of following functions. Is the inverse a function? State the domain and range for each inverse.

a) $f(x) = \{(-2, 4), (1, 3), (4, 7), (8, 11), (10, 4)\}$



Ex. 2: Draw the inverse of the function below. Is the inverse a function? State the domain and range for the inverse.



To find the equation of the inverse of a function,

- i) Replace $f(x)$ with y .
- ii) Switch the x and y variables in the equation.
- iii) Solve for y .
- iv) If the inverse is a function, then replace y with $f^{-1}(x)$. If the inverse is a relation, leave the equation with y .

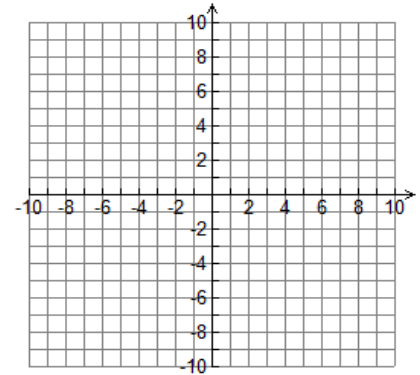
Ex. 3: Determine the equation of the inverse of the following functions.

a) $f(x) = 2 - 5x$

b) $h(x) = \sqrt{-x + 4} - 7$

c) $g(x) = 2(x - 1)^2 + 3$

Ex. 4: The inverse of 3c) is not a function. What can we do to $g(x)$ so that its inverse will be a function?



Practice Questions

1. Determine the inverse relation for each set of ordered pairs. Which of the relations and inverse relations are functions?

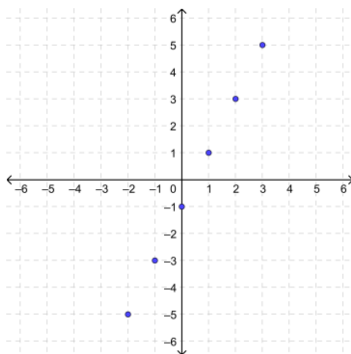
a) $\{(-2, 3), (0, 4), (2, 5), (4, 6)\}$

b) $\{(2, 5), (2, -1), (3, 1), (5, 1)\}$

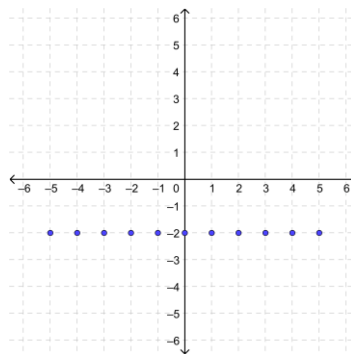
c) $\{(-3, -4), (-2, 1), (-1, 4), (0, 5), (1, 4), (2, 1), (3, -4)\}$

2. Graph the inverse for each of the following functions. State the domain and range for each function and its inverse. Which inverse relations are functions?

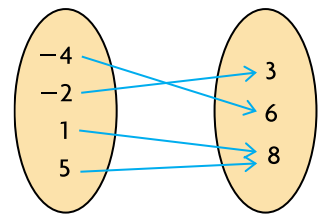
a)



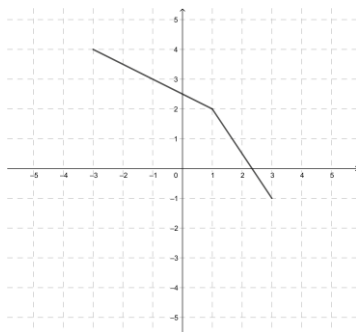
b)



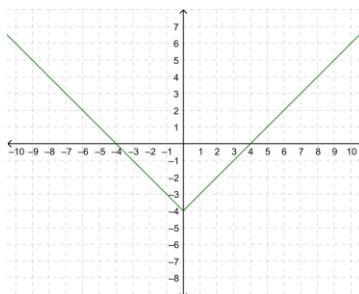
c)



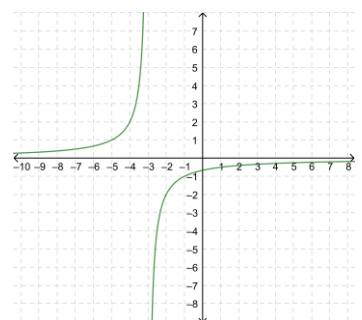
d)

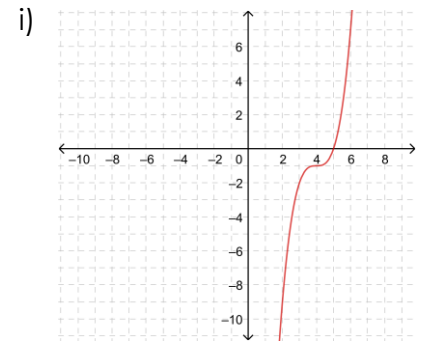
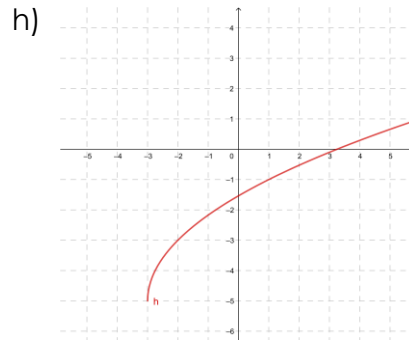
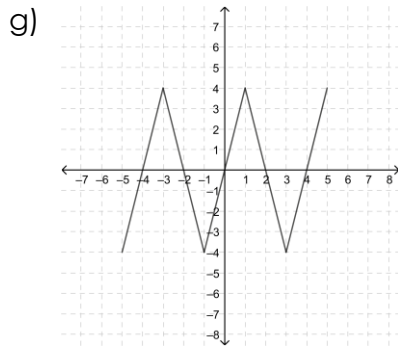


e)



f)





3. a) Determine $f^{-1}(x)$ for the linear function $f(x) = 5x - 1$.
 b) Graph f and f^{-1} on the same axes.
 c) Explain how you can tell that f^{-1} is also a linear function.
 d) State the coordinates of any points that are common to both f and f^{-1} .
 e) Compare the slopes of the two lines.
 f) Repeat parts a) to e) for $g(x) = -\frac{1}{2}x + 3$, $h(x) = 6 - x$, and $p(x) = 4$.
4. The ordered pair $(1, 5)$ belongs to a function f . Explain why the ordered pair $(2, 1)$ cannot belong to f^{-1} .
5. Given $f(x) = k(2 + x)$, find the value of k if $f^{-1}(-2) = -3$.
6. Determine the inverse of each of the functions below.
 a) $f(x) = -(x + 3)^2 - 7$ b) $g(x) = \frac{1}{4}(5x - 1)^2$ c) $h(x) = 2(7 - x)^2 + 9$
7. a) Graph $f(x) = 3(x - 2)^2 - 5$. State the domain and range.
 b) Graph $f^{-1}(x)$. State the domain and range.
 c) Is f^{-1} function? State the domain for f so that f^{-1} is a function.
8. Given $f(x) = 7 - 2(x - 1)^2$, $x \geq 1$, determine
 a) $f(3)$ b) $f^{-1}(x)$ c) $f^{-1}(5)$ d) $f^{-1}(2a + 7)$
9. a) Graph $f(x) = -\sqrt{x + 4} - 5$. State the domain and range.
 b) Graph $f^{-1}(x)$. State the domain and range. Is f^{-1} function?
 c) Determine the equation for f^{-1} .
10. Self-inverse functions are their own inverses. Find three functions that are self-inverse.
11. Determine the inverse of each function.
 a) $f(x) = \frac{2}{x} - 5$ b) $f(x) = \frac{-1}{x + 3} + 8$

Is it a FUNCTION?

	Yes	No
table		
mapping		
graph		

Evaluating FUNCTIONS

the function of x
 $f(x)$ is read as "f of x"

If $f(x) = x^2 + 4x$,

find $f(2)$:

$$f(x) = x^2 + 4x$$

$$f() =$$

$$f() =$$

$$f() =$$

$$f() =$$

find $f(-3)$:

$$f(x) = x^2 + 4x$$

$$f() =$$

$$f() =$$

$$f() =$$

$$f() =$$

Functions!

Cubic: $f(x) = x^3$
 (x, y)

Quadratic: $f(x) = x^2$
 (x, y)

Root: $f(x) = \sqrt{x}$
 (x, y)

RECIPROCAL: $f(x) = \frac{1}{x}$
 (x, y)

Absolute: $f(x) = |x|$
 (x, y)

Domain & Range

The DOMAIN is the set of all
 for a _____.

The RANGE is the set of all
 for a _____.

Transformations

$$g(x) = a[k(x-d)] + c$$

a - _____
 k - _____
 d - _____
 c - _____

Mapping Notation: (x , y)

Finding the INVERSE Function

Original function: $f(x) = 3x + 2$

① Replace $f(x)$
 with

② the
 and values

③ Solve for

④ Replace
 with

The INVERSE Function!

Relax ... Review

- Describe the graphs of the relations, $y = 3$ and $x = 3$. Are these relations functions? Explain.
- Determine which of the following relations are functions. Explain.
 a) $y = \sqrt{x+2}$ b) $y = 2 - x$ c) $y = 2x^2 - 3$
- Determine the domain and range of each relation in question 2.
- Determine the domain and range of a circle with centre (2, 5) and radius of 4.
- Determine the domain and range of a parabola with vertex (4, 3) and $y = 3$ its minimum value.
- If $f(x) = 3(x-1)^2 - 4$, determine
 a) $f(-1)$ b) $f(2) - f(1)$ c) $f(1-x)$
- Consider the functions, $f(x) = x^2 - 3x$ and $g(x) = 1 - 2x$. Determine $f(c+2) - g(c+2)$.
- If $f(x+1) = \frac{2f(x)+1}{3}$ and $f(2) = 2$, find the value of $f(1)$.
- Let $f(x) = 3x+1$ and $g(x) = 2-x$. Determine the values of a such that:
 a) $f(a) = g(a)$ b) $f(a^2) = g(2a)$
- State the mapping notation for the following.
 a) $y = \frac{1}{3}f(-x+4) - 5$ b) $y = -3f(2x+3)+8$
- Suppose (-2, 9) is a point on the graph of $y = f(x)$. Give the coordinates of a point on the graph of:
 a) $y = 2f(x)$ b) $y = f(2x)$ c) $y = f(x-5)$ d) $y = f(x)+4$
 e) $y = -f(x+1) - 3$
- Describe the following transformations.
 a) Given $y = x^2$, describe the graph of $y = 2f(-x)+5$.
 b) Given $y = \sqrt{x}$, describe the graph of $y = \frac{1}{4}f(3x-6)$.
- For each of the following, determine the equation of the new function.
 a) The graph $f(x) = \sqrt{x}$ is reflected across the y-axis, compressed vertically by a factor of $\frac{1}{3}$, and then translated 5 units right and 2 units down.
 b) The graph $f(x) = x^2$ is vertically stretched by a factor of 2, reflected in the x-axis and translated 4 units right.

c) The graph $f(x) = \frac{1}{x}$ is vertically compressed by a factor of $\frac{1}{2}$, reflected in the y-axis and then translated 3 units left and 4 units down.

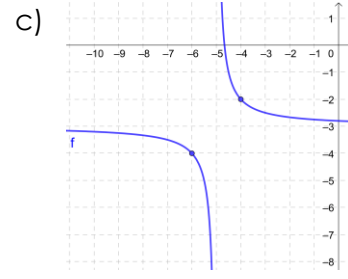
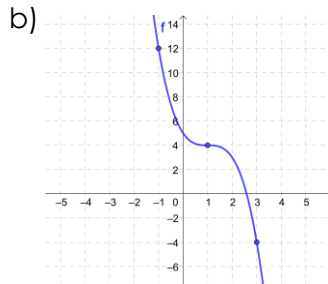
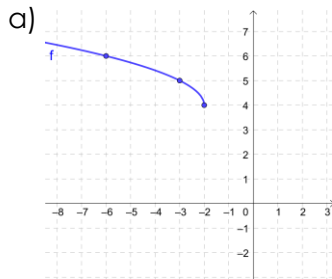
14. Without sketching the graph, determine the domain and range for each of the functions below.

a) $f(x) = \frac{5}{x+2} - 8$

b) $f(x) = 3\sqrt{x-4} + 1$

c) $f(x) = -(2x-4)^2 + 5$

15. Given the following graphs that have gone through transformations, identify the parent function and write the equation of the transformed function for each.



16. For each of the following functions, describe the transformations and then graph. State the domain and range for each.

a) $f(x) = \sqrt{-2x-8}$

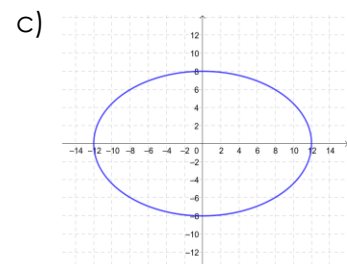
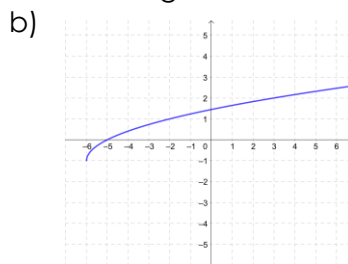
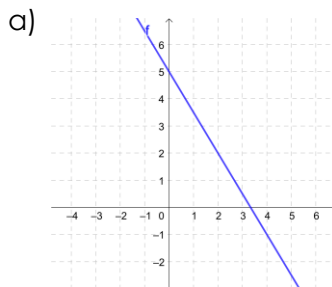
b) $g(x) = \frac{1}{2}(x-4)^3 - 5$

c) $h(x) = \frac{2}{3x} + 7$

d) $f(x) = -2|x-1| - 3$

17. Explain how a relation and its inverse are related. Provide a linear example.

18. Sketch the inverse of each of the following relations.



19. Given $f(x) = \frac{1}{3}(x-2)^2 - 6$,

a) determine the equation of the inverse.

b) graph $f(x)$ and its inverse.

c) restrict the domain of $f(x)$ so that the inverse is also a function.

20. Given $f(x) = k(3-x)$, find the value of k if $f^{-1}(9) = -6$.